

Support Vector Machines

COMP0078: Supervised Learning

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Revision notation: items marked with $\star$ are specifically listed in the revision notes file ([revision25-26-V1.pdf](#)). All content in this file is examinable.

1 Optimal Separating Hyperplane (Linearly Separable Case) $\star$

1.1 Hyperplanes and Margins

Let $S = \{(\mathbf{x}_i, y_i)\}_{i=1}^m \in \mathbb{R}^n \times \{-1, 1\}$ be a training set. We assume initially that the data are linearly separable.

Definition 1.1 (Separating Hyperplane $\star$). A hyperplane is a set $H_{\mathbf{w},b} = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{w}^\top \mathbf{x} + b = 0\}$, parameterized by $\mathbf{w} \in \mathbb{R}^n$ and $b \in \mathbb{R}$. It is called a **separating hyperplane** if there exist $\mathbf{w}$ and b such that:

$$y_i(\mathbf{w}^\top \mathbf{x}_i + b) > 0, \quad i = 1, \dots, m. \quad (1)$$

Note that we require the inequality in (1) to be strict (we do not admit that data lie on the hyperplane).

Definition 1.2 (Geometric Margin $\star$). The distance $\rho_{\mathbf{x}}(\mathbf{w}, b)$ of a point $\mathbf{x}$ from a hyperplane $H_{\mathbf{w},b}$ (Definition 1.1) is given by:

$$\rho_{\mathbf{x}}(\mathbf{w}, b) := \frac{|\mathbf{w}^\top \mathbf{x} + b|}{\|\mathbf{w}\|}. \quad (2)$$

If $H_{\mathbf{w},b}$ separates the training set S , we define the **margin** of the set S as:

$$\rho_S(\mathbf{w}, b) := \min_{i \in [m]} \rho_{\mathbf{x}_i}(\mathbf{w}, b). \quad (3)$$

We also define the **signed margin** of a single point $\mathbf{x}$ as $\frac{\mathbf{w}^\top \mathbf{x} + b}{\|\mathbf{w}\|}$ (note that this can be positive or negative).

Remark 1.3 (Projection Verification). To verify the distance formula (2), consider the projection $\mathbf{p}$ of $\mathbf{x}$ onto $H_{\mathbf{w},b}$:

$$\mathbf{p} = \mathbf{x} - \frac{\mathbf{w}(b + \langle \mathbf{w}, \mathbf{x} \rangle)}{\|\mathbf{w}\|^2}.$$

One verifies that: (i) $\mathbf{p}$ lies on the hyperplane, since $\langle \mathbf{w}, \mathbf{p} \rangle + b = \langle \mathbf{w}, \mathbf{x} \rangle - \frac{\|\mathbf{w}\|^2(b + \langle \mathbf{w}, \mathbf{x} \rangle)}{\|\mathbf{w}\|^2} + b = 0$; and (ii) the vector $\mathbf{x} - \mathbf{p}$ is orthogonal to the hyperplane. The distance is then:

$$\rho_{\mathbf{x}}(\mathbf{w}, b) = \|\mathbf{x} - \mathbf{p}\| = \frac{|b + \langle \mathbf{w}, \mathbf{x} \rangle|}{\|\mathbf{w}\|}.$$

1.2 Canonical Parameterization

The pair $(\mathbf{w}, b)$ is not unique; rescaling by a positive constant results in the same hyperplane. We typically use the **Canonical Hyperplane** parameterization.

Definition 1.4 (Canonical Hyperplane*). *We choose $\|\mathbf{w}\|$ such that the point closest to the hyperplane has a functional margin of 1. That is, we require:*

$$\min_{i \in [m]} y_i(\mathbf{w}^\top \mathbf{x}_i + b) = 1. \quad (4)$$

Under this parameterization, the geometric margin (Definition 1.2) becomes:

$$\rho_S(\mathbf{w}, b) = \frac{1}{\|\mathbf{w}\|}. \quad (5)$$

Remark 1.5 (Normalized vs. Canonical Hyperplane*). *An alternative parameterization is the **normalized hyperplane** where $\|\mathbf{w}\| = 1$, in which case $\rho_{\mathbf{x}}(\mathbf{w}, b) = |\mathbf{w}^\top \mathbf{x} + b|$ and $\rho_S(\mathbf{w}, b) = \min_{i \in [m]} y_i(\mathbf{w}^\top \mathbf{x}_i + b)$. In the canonical parameterization (Definition 1.4), we instead have a data-dependent normalization satisfying (4). We will mainly work with the canonical parameterization.*

1.3 The Primal Optimization Problem

The Optimal Separating Hyperplane (OSH) is the one that maximizes the margin. Using the canonical parameterization (Definition 1.4), maximizing $\frac{1}{\|\mathbf{w}\|}$ (i.e. ρ_S from (5)) is equivalent to minimizing $\|\mathbf{w}\|^2$.

Proposition 1.6 (Hard Margin Primal – P1*). *The OSH is found by solving:*

$$\begin{aligned} & \text{Minimize} \quad \frac{1}{2} \mathbf{w}^\top \mathbf{w} \\ & \text{subject to} \quad y_i(\mathbf{w}^\top \mathbf{x}_i + b) \geq 1, \quad i = 1, \dots, m. \end{aligned} \quad (6)$$

This is a convex quadratic programming (QP) problem. The quantity $1/\|\mathbf{w}\|$ is the margin of the OSH.

2 Solving via the Lagrangian*

To solve the constrained optimization in Proposition 1.6, we employ the method of Lagrange multipliers.

2.1 General Constrained Optimisation

Remark 2.1 (Lagrangian Duality for Convex Problems*). *Consider minimising a convex function $f(\mathbf{z})$ subject to linear inequality constraints $A\mathbf{z} \leq \mathbf{c}$, where $f: \mathbb{R}^n \rightarrow \mathbb{R}$ is differentiable, A is $m \times n$, and $\mathbf{c} \in \mathbb{R}^m$. If the feasible set $\{\mathbf{z} : A\mathbf{z} \leq \mathbf{c}\}$ is non-empty, we form the Lagrangian*

$$L(\mathbf{z}, \boldsymbol{\alpha}) := f(\mathbf{z}) + \boldsymbol{\alpha}^\top (A\mathbf{z} - \mathbf{c}),$$

and we have $\max_{\boldsymbol{\alpha} \geq 0} \min_{\mathbf{z}} L(\mathbf{z}, \boldsymbol{\alpha}) = \min_{\mathbf{z}} \{f(\mathbf{z}) : A\mathbf{z} \leq \mathbf{c}\}$. The necessary and sufficient conditions (KKT) for a solution $(\bar{\boldsymbol{\alpha}}, \bar{\mathbf{z}})$ are:

1. $A\bar{\mathbf{z}} \leq \mathbf{c}$ (primal feasibility),
2. $\bar{\boldsymbol{\alpha}} \geq 0$ (dual feasibility),
3. $\nabla_{\mathbf{z}} L|_{\bar{\mathbf{z}}} = 0$ (stationarity),
4. $(A\bar{\mathbf{z}} - \mathbf{c})_i \bar{\alpha}_i = 0, i = 1, \dots, m$ (complementary slackness).

2.2 The Lagrangian and Saddle Point

We define the Lagrangian function for the primal problem (6) (Proposition 1.6), where $\alpha_i \geq 0$ are the Lagrange multipliers:

$$L(\mathbf{w}, b; \boldsymbol{\alpha}) = \frac{1}{2} \mathbf{w}^\top \mathbf{w} - \sum_{i=1}^m \alpha_i [y_i (\mathbf{w}^\top \mathbf{x}_i + b) - 1]. \quad (7)$$

We minimize L with respect to the primal variables $(\mathbf{w}, b)$ and maximize with respect to the dual variables $\boldsymbol{\alpha}$ (cf. Remark 2.1).

Differentiating with respect to $\mathbf{w}$ and b and setting to zero:

$$\frac{\partial L}{\partial b} = - \sum_{i=1}^m y_i \alpha_i = 0, \quad (8)$$

$$\frac{\partial L}{\partial \mathbf{w}} = \mathbf{w} - \sum_{i=1}^m \alpha_i y_i \mathbf{x}_i = 0 \implies \mathbf{w} = \sum_{i=1}^m \alpha_i y_i \mathbf{x}_i. \quad (9)$$

2.3 The Dual Problem

Substituting (9) and using (8) back into the Lagrangian (7) leads to the Dual Problem.

Proposition 2.2 (Hard Margin Dual – P2*). *Let $\mathbf{A}$ be the $m \times m$ Gram matrix where $A_{ij} = y_i y_j \mathbf{x}_i^\top \mathbf{x}_j$. The dual problem is:*

$$\begin{aligned} \text{Maximize} \quad & Q(\boldsymbol{\alpha}) := \sum_{i=1}^m \alpha_i - \frac{1}{2} \boldsymbol{\alpha}^\top \mathbf{A} \boldsymbol{\alpha} \\ \text{subject to} \quad & \sum_{i=1}^m y_i \alpha_i = 0, \\ & \alpha_i \geq 0, \quad i = 1, \dots, m. \end{aligned} \quad (10)$$

Note that the complexity of the dual problem depends on the number of samples m , not the input dimension n (analogous to ridge regression).

2.4 KKT Conditions and Support Vectors

If $\bar{\boldsymbol{\alpha}}$ is a solution to the dual (Proposition 2.2), the primal solution is given by the representation (9):

$$\bar{\mathbf{w}} = \sum_{i=1}^m \bar{\alpha}_i y_i \mathbf{x}_i. \quad (11)$$

The solution satisfies the KKT complementary slackness condition (cf. Remark 2.1):

$$\bar{\alpha}_i [y_i (\bar{\mathbf{w}}^\top \mathbf{x}_i + \bar{b}) - 1] = 0, \quad i = 1, \dots, m. \quad (12)$$

Definition 2.3 (Support Vectors*). *The input vectors $\mathbf{x}_i$ for which $\bar{\alpha}_i > 0$ are called **Support Vectors (SVs)**.*

Remark 2.4 (Properties of Support Vectors). • *The OSH is determined only by the support vectors (Definition 2.3). In particular, $\bar{\mathbf{w}}$ in (11) is a linear combination of only the $\mathbf{x}_i$ for which $\bar{\alpha}_i > 0$.*

- *Removing non-support vectors and retraining yields the same hyperplane.*

- For any support vector $\mathbf{x}_j$, by complementary slackness (12) we have $y_j(\bar{\mathbf{w}}^\top \mathbf{x}_j + \bar{b}) = 1$. This allows us to compute the bias: $\bar{b} = y_j - \bar{\mathbf{w}}^\top \mathbf{x}_j$.
- A new point $\mathbf{x}$ is classified as $\text{sgn}(\sum_{i=1}^m y_i \bar{\alpha}_i \mathbf{x}_i^\top \mathbf{x} + \bar{b})$.

Theorem 2.5 (SV Generalization Bound^{*}). Let n_{sv} be the expected number of support vectors (Definition 2.3) for an SVM trained on m examples sampled i.i.d. The expected generalization error of an SVM trained on $m-1$ examples is bounded above by $\frac{n_{\text{sv}}}{m}$. This suggests that sparsity in SVs implies better generalization.

2.5 Intuitions on Margin and Sparsity

Remark 2.6 (Margin, Sparsity, and Generalization^{*}). Several key intuitions connect the margin, support vector sparsity, and generalization:

1. **Sparsity of SVs implies generalization:** By Theorem 2.5, fewer support vectors yield a tighter generalization bound.
2. **Large margin implies generalization:** There exist formal bounds in the literature relating the margin to the generalization error; these are beyond the scope of this course.
3. **Larger margin implies fewer SVs:** All else being equal, a larger margin tends to yield fewer support vectors. This can be understood via the convex hull interpretation: the SVM optimisation is equivalent to finding the shortest line segment connecting the convex hull of the positive points to that of the negative points; the perpendicular bisecting hyperplane of this segment is the classifier.
4. **Face dimension and SV count:** The number of support vectors associated with a face of the convex hull equals the number of vertices of that face. A point nearer to a convex polytope tends to be closest to a higher-dimensional face, involving more vertices (SVs).

3 Soft Margin SVM (Linearly Non-Separable Case)^{*}

3.1 Hinge Loss and Slack Variables

If data is not linearly separable, the hard margin primal (Proposition 1.6) is infeasible. We introduce **slack variables** $\xi_i \geq 0$ to allow for margin violations. Ideally, we would minimize the misclassification loss $V_{\text{mc}}(y, \hat{y}) = \mathcal{I}[y \neq \text{sign}(\hat{y})]$, but this is NP-hard. Instead, we convexify by using the **Hinge Loss**:

Definition 3.1 (Hinge Loss). The hinge loss is defined as:

$$V_{\text{hinge}}(y, \hat{y}) = \max(0, 1 - y\hat{y}). \quad (13)$$

Proposition 3.2 (Soft Margin Primal – P3^{*}). The soft margin SVM solves:

$$\begin{aligned} & \text{Minimize} \quad \frac{1}{2} \mathbf{w}^\top \mathbf{w} + C \sum_{i=1}^m \xi_i \\ & \text{subject to} \quad y_i(\mathbf{w}^\top \mathbf{x}_i + b) \geq 1 - \xi_i, \\ & \quad \quad \quad \xi_i \geq 0, \quad i = 1, \dots, m. \end{aligned} \quad (14)$$

The parameter $C > 0$ controls the trade-off between maximizing the margin ($\|\mathbf{w}\|^{-1}$) and minimizing the training error ($\sum_i \xi_i$). The slack variables ξ_i relax the separation constraints: $\xi_i > 0 \Rightarrow y_i(\mathbf{w}^\top \mathbf{x}_i + b) < 1$, i.e. the margin is less than 1.

3.2 Soft Margin Lagrangian and Dual

The Lagrangian for the soft margin primal (14) (Proposition 3.2) includes multipliers $\alpha_i \geq 0$ (for the separation constraints) and $\beta_i \geq 0$ (for non-negativity of ξ_i):

$$L(\mathbf{w}, \boldsymbol{\xi}, b; \boldsymbol{\alpha}, \boldsymbol{\beta}) = \frac{1}{2} \mathbf{w}^\top \mathbf{w} + C \sum_{i=1}^m \xi_i - \sum_{i=1}^m \alpha_i [y_i(\mathbf{w}^\top \mathbf{x}_i + b) + \xi_i - 1] - \sum_{i=1}^m \beta_i \xi_i. \quad (15)$$

We minimize over $(\mathbf{w}, \boldsymbol{\xi}, b)$ and maximize over $\boldsymbol{\alpha}, \boldsymbol{\beta} \geq 0$. Differentiating:

$$\frac{\partial L}{\partial b} = - \sum_{i=1}^m y_i \alpha_i = 0, \quad (16)$$

$$\frac{\partial L}{\partial \mathbf{w}} = \mathbf{w} - \sum_{i=1}^m \alpha_i y_i \mathbf{x}_i = 0 \implies \mathbf{w} = \sum_{i=1}^m \alpha_i y_i \mathbf{x}_i, \quad (17)$$

$$\frac{\partial L}{\partial \xi_i} = C - \alpha_i - \beta_i = 0 \implies 0 \leq \alpha_i \leq C. \quad (18)$$

Substituting (17)–(18) into the Lagrangian (15) yields the dual problem, analogous to the derivation of Proposition 2.2.

Proposition 3.3 (Soft Margin Dual – P4*).

$$\begin{aligned} \text{Maximize } Q(\boldsymbol{\alpha}) &:= \sum_{i=1}^m \alpha_i - \frac{1}{2} \boldsymbol{\alpha}^\top \mathbf{A} \boldsymbol{\alpha} \\ \text{subject to } &\sum_{i=1}^m y_i \alpha_i = 0, \\ &0 \leq \alpha_i \leq C, \quad i = 1, \dots, m. \end{aligned} \quad (19)$$

This is identical to the hard margin dual (Proposition 2.2) except that now we have “box constraints” $0 \leq \alpha_i \leq C$. If the data is linearly separable, choosing C large enough recovers the OSH from Proposition 1.6.

3.3 KKT Conditions for the Soft Margin

The KKT complementary slackness conditions for the soft margin problem (Proposition 3.2) are:

$$\bar{\alpha}_i [y_i(\bar{\mathbf{w}}^\top \mathbf{x}_i + \bar{b}) - 1 + \bar{\xi}_i] = 0, \quad (20)$$

$$(C - \bar{\alpha}_i) \bar{\xi}_i = 0. \quad (21)$$

Note that (21) follows since $\beta_i = C - \alpha_i$ (from (18)). Again, points for which $\bar{\alpha}_i > 0$ are termed support vectors (Definition 2.3).

Remark 3.4 (KKT Classification of Points). Equations (20) and (21) imply the following trichotomy:

1. $y_i(\bar{\mathbf{w}}^\top \mathbf{x}_i + \bar{b}) > 1 \implies \bar{\alpha}_i = 0$ (correctly classified, not a SV).
2. $y_i(\bar{\mathbf{w}}^\top \mathbf{x}_i + \bar{b}) = 1 \implies \bar{\alpha}_i \in [0, C]$ (SV on the margin, $\bar{\xi}_i = 0$).
3. $y_i(\bar{\mathbf{w}}^\top \mathbf{x}_i + \bar{b}) < 1 \implies \bar{\alpha}_i = C$ (SV with positive slack $\bar{\xi}_i > 0$, potentially misclassified).

Conversely: if $\bar{\alpha}_i = 0$ then $y_i(\bar{\mathbf{w}}^\top \mathbf{x}_i + \bar{b}) \geq 1, \bar{\xi}_i = 0$; if $\bar{\alpha}_i \in (0, C)$ then $y_i(\bar{\mathbf{w}}^\top \mathbf{x}_i + \bar{b}) = 1, \bar{\xi}_i = 0$; if $\bar{\alpha}_i = C$ then $y_i(\bar{\mathbf{w}}^\top \mathbf{x}_i + \bar{b}) \leq 1, \bar{\xi}_i \geq 0$.

3.4 The Role of the Parameter C^*

Remark 3.5 (Trade-Off Controlled by C^*). *The parameter C in Proposition 3.2 controls the trade-off between $\|\mathbf{w}\|^2$ (margin size) and the training error $\sum_{i=1}^m \xi_i$. Both the margin and the training error are non-increasing functions of C . It can be shown that the optimal Lagrange multipliers $\bar{\alpha}_i$ (and hence $\bar{\mathbf{w}}, \bar{b}$) are piecewise quadratic functions of C .*

Selecting C :

- C is often selected by minimizing the leave-one-out (LOO) cross-validation error.
- We need to “retrain” the SVM no more than n_{sv} times (since removing a non-SV does not change the solution, by Remark 2.4).
- Alternatively, one may use $\frac{n_{\text{sv}}}{m}$ as an upper bound on the LOO CV error (by Theorem 2.5).

4 Kernels and Regularization

4.1 Kernel SVM

The analysis above holds if we work with a feature map $\phi : \mathcal{X} \rightarrow \mathcal{W}$. We simply replace $\mathbf{x}$ by $\phi(\mathbf{x})$ and $\mathbf{x}_i^\top \mathbf{x}_j$ by $\langle \phi(\mathbf{x}_i), \phi(\mathbf{x}_j) \rangle$.

Definition 4.1 (Kernel Function). *A kernel function $K : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ is defined as:*

$$K(\mathbf{x}_i, \mathbf{x}_j) = \langle \phi(\mathbf{x}_i), \phi(\mathbf{x}_j) \rangle. \quad (22)$$

Notice that the dual problem (Proposition 3.3) depends on the data only via the Gram matrix $A_{ij} = y_i y_j \mathbf{x}_i^\top \mathbf{x}_j$, which is now replaced by $A_{ij} = y_i y_j K(\mathbf{x}_i, \mathbf{x}_j)$.

Definition 4.2 (SVM Decision Function). *An SVM with kernel K (Definition 4.1) classifies a new point $\mathbf{x} \in \mathcal{X}$ via:*

$$f(\mathbf{x}) = \sum_{i=1}^m y_i \bar{\alpha}_i K(\mathbf{x}_i, \mathbf{x}) + \bar{b}, \quad (23)$$

where $\bar{\alpha}_i$ solve the dual problem (Proposition 3.3) with kernelized Gram matrix, and $\bar{b}$ is obtained from the KKT conditions (Remark 3.4). A new point is classified as $\text{sgn}(f(\mathbf{x}))$.

4.2 Connection to Regularization*

The Soft Margin SVM formulation (Proposition 3.2) is equivalent to Regularized Empirical Risk Minimization with the hinge loss (Definition 3.1).

Proposition 4.3 (SVM as Regularized ERM*). *The soft margin primal (14) is equivalent to:*

$$\min_{\mathbf{w}, b} E_\lambda(\mathbf{w}, b) := \sum_{i=1}^m \max(0, 1 - y_i(\langle \mathbf{w}, \phi(\mathbf{x}_i) \rangle + b)) + \lambda \|\mathbf{w}\|^2 \quad (24)$$

where $\lambda = \frac{1}{2C}$.

Proof. We show that problems (14) and (24) have the same minimizer (up to the constant C). Starting from the soft margin primal (Proposition 3.2):

$$\begin{aligned}
& \min_{\mathbf{w}, b, \xi} \left\{ C \sum_{i=1}^m \xi_i + \frac{1}{2} \|\mathbf{w}\|^2 : y_i(\langle \mathbf{w}, \phi(\mathbf{x}_i) \rangle + b) \geq 1 - \xi_i, \xi_i \geq 0 \right\} \\
&= \min_{\mathbf{w}, b} \left\{ \min_{\xi} \left\{ C \sum_{i=1}^m \xi_i + \frac{1}{2} \|\mathbf{w}\|^2 : \xi_i \geq 1 - y_i(\langle \mathbf{w}, \phi(\mathbf{x}_i) \rangle + b), \xi_i \geq 0 \right\} \right\} \\
&= \min_{\mathbf{w}, b} \left\{ C \sum_{i=1}^m \max(1 - y_i(\langle \mathbf{w}, \phi(\mathbf{x}_i) \rangle + b), 0) + \frac{1}{2} \|\mathbf{w}\|^2 \right\} \\
&= C \cdot E_{1/(2C)}(\mathbf{w}, b).
\end{aligned}$$

□

This highlights that SVM minimizes the Hinge Loss (Definition 3.1) with L_2 regularization.

5 SVM Regression

Remark 5.1 (SVM for Regression). *SVMs can also be developed for regression using the ε -insensitive loss $|y - f(\mathbf{x})|_\varepsilon = \max(|y - f(\mathbf{x})| - \varepsilon, 0)$. The optimization problem becomes:*

$$\begin{aligned}
& \text{Minimize} \quad \frac{1}{2} \mathbf{w}^\top \mathbf{w} + C \sum_{i=1}^m (\xi_i + \xi_i^*) \\
& \text{subject to} \quad \mathbf{w}^\top \mathbf{x}_i + b - y_i \leq \varepsilon + \xi_i, \\
& \quad \quad \quad y_i - \mathbf{w}^\top \mathbf{x}_i - b \leq \varepsilon + \xi_i^*, \\
& \quad \quad \quad \xi_i, \xi_i^* \geq 0, \quad i = 1, \dots, m.
\end{aligned} \tag{25}$$

Both the classification (Definition 3.1) and regression loss functions are scale sensitive: errors below a certain resolution do not count. This leads to sparse solutions.

6 Solution Methods

For general kernels, we must solve the dual (Proposition 3.3).

- For small m , standard QP solvers (e.g. Interior Point methods) work well.
- For large m (e.g., $m > 10^5$), the dense $m \times m$ matrix $\mathbf{A}$ makes standard QP impractical.
- **Decomposition Methods (e.g., SMO):** Iteratively optimize a small subset (active set $\mathcal{A}$) of $q \leq m$ variables while keeping others fixed. At each iteration, one removes a variable satisfying the KKT conditions and adds one violating them. One can show that $Q(\alpha)$ increases after each iteration.

If $m \ll N$ (number of features), it may be more efficient to solve the primal problem directly.

7 Problems and Exercises

Problem 1. Consider the optimization given by:

$$\min_{\mathbf{w}, b, \xi} \|\mathbf{w}\|^2 + C \sum_{i=1}^m \xi_i$$

subject to $y_i \langle \mathbf{w}, \mathbf{x}_i \rangle \geq 1 - \xi_i$ and $\xi_i \geq 0$. Identify the regularizer and the loss function.

Solution Sketch. The term $\|\mathbf{w}\|^2$ corresponds to the regularizer (specifically L_2 regularization), ensuring the margin is maximized (cf. Proposition 4.3). The term $\sum_i \xi_i$ represents the empirical risk using the Hinge Loss (Definition 3.1): $L(y, f(\mathbf{x})) = \max(0, 1 - yf(\mathbf{x}))$. $\square$

Problem 2. Show that the vector $\mathbf{w}$ solving the hard-margin Primal (Proposition 1.6) has the form $\mathbf{w} = \sum_{i=1}^m c_i y_i \mathbf{x}_i$ for non-negative coefficients c_i .

Solution Sketch. This follows directly from the Lagrangian stationarity condition (9): $\frac{\partial L}{\partial \mathbf{w}} = 0$ gives $\mathbf{w} = \sum_{i=1}^m \alpha_i y_i \mathbf{x}_i$, where $c_i := \alpha_i \geq 0$ are the Lagrange multipliers. $\square$

Problem 3. Show that the coefficients $c_1, \dots, c_m$ from Problem 2 solve the dual optimization:

$$\max \left\{ -\frac{1}{2} \sum_{i,j=1}^m c_i c_j y_i y_j \mathbf{x}_i^\top \mathbf{x}_j + \sum_{i=1}^m c_i : c_j \geq 0, j = 1, \dots, m \right\}.$$

Furthermore, if $(\hat{c}_1, \dots, \hat{c}_m)$ is the solution and $\hat{\mathbf{w}}$ is the solution to Proposition 1.6, argue that $\hat{\mathbf{w}}^\top \hat{\mathbf{w}} = \sum_{i=1}^m \hat{c}_i$.

Solution Sketch. Substituting $\mathbf{w} = \sum_i c_i y_i \mathbf{x}_i$ (from Problem 2) into the Lagrangian (7) and using the stationarity conditions (8)–(9) yields the dual (Proposition 2.2, noting $b = 0$ since the hyperplane passes through the origin). For the final identity: at the optimum, $\hat{\mathbf{w}}^\top \hat{\mathbf{w}} = 2 \cdot \frac{1}{2} \hat{\mathbf{w}}^\top \hat{\mathbf{w}} = 2Q(\hat{\mathbf{c}}) - 2 \sum_i \hat{c}_i + \hat{\mathbf{w}}^\top \hat{\mathbf{w}}$; using strong duality and the dual objective at optimum, one obtains $\hat{\mathbf{w}}^\top \hat{\mathbf{w}} = \sum_i \hat{c}_i$. $\square$